



ULUSLARARASI KIBRIS ÜNİVERSİTESİ
CYPRUS INTERNATIONAL UNIVERSITY

CYPRUS INTERNATIONAL UNIVERSITY

FACULTY of ENGINEERING

**DATABASE MANAGEMENT SYSTEMS AND
PROGRAMMING II**

STRAWBERRY FARMING CORP.

Students and Roles ;

- **Fatmatül Zehra Taş (22208075):** DDL (table creation, constraints), DML (insert, update, delete operations), database connection (Npgsql), SQL queries, and User Interface (UI) development
- **Hediye Sultan Bozkurt (22319266):** Report writing, PL/pgSQL (procedures, functions, triggers), and advanced SQL queries
- **Zeynep Özlüer (22315460):** ER Diagram (ERD) design, database modeling, and contribution to SQL queries.
- **Emre Özel (22315566):** User Interface (UI) design (WinForms), application layout, and database integration

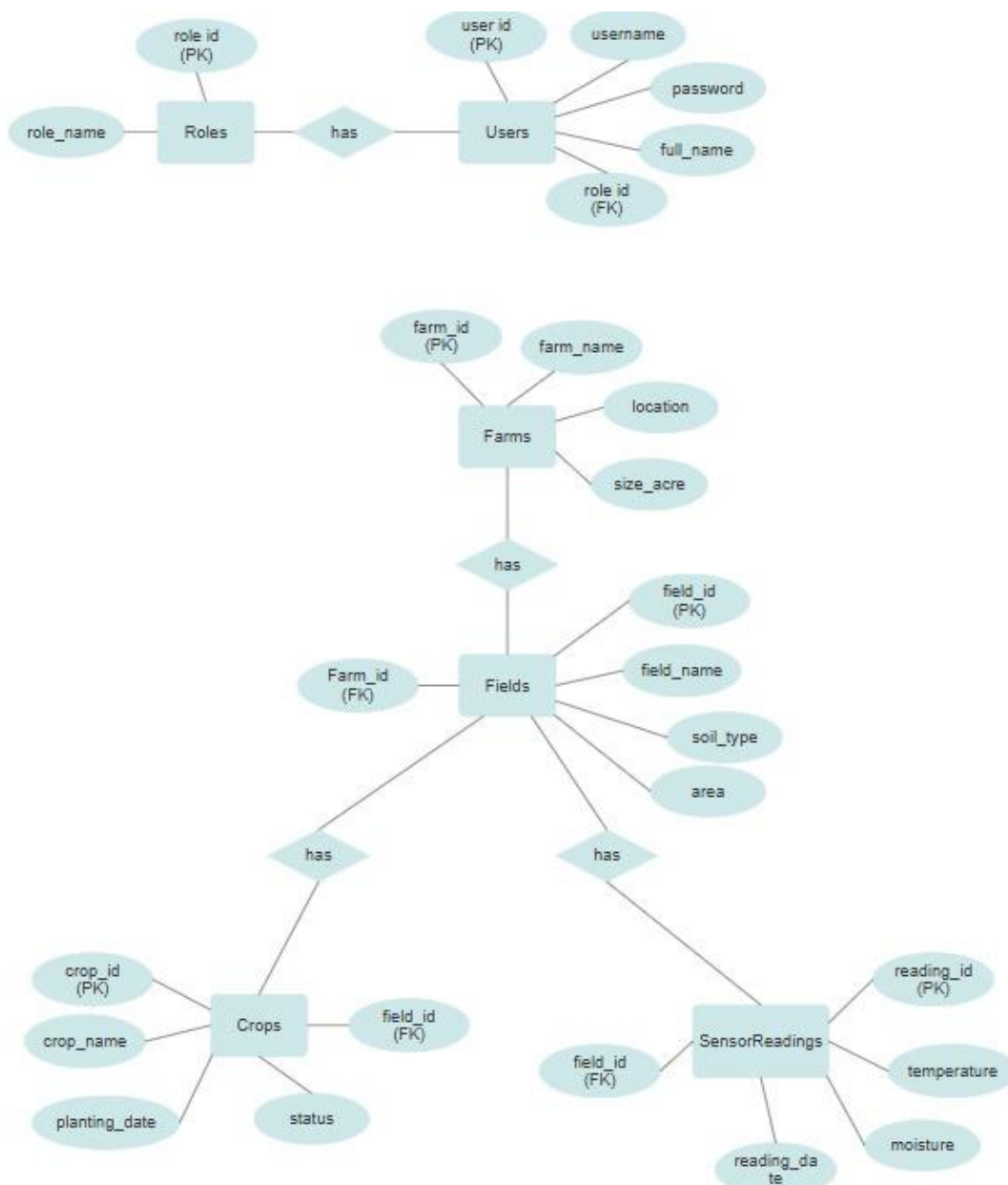
Instructor: *PROF. DR. MELİKE ŞAH DİREKOĞLU*

Introduction

This project presents a Smart Farming Database Management System designed to manage farms, fields, crops, and sensor data such as temperature and moisture. The system allows users to store, update, and analyze data through a user-friendly interface.

It assumes that each farm contains multiple fields, and each field has crops and sensor readings. The system helps monitor conditions and detect risks, supporting better decision-making in agriculture.

1) ER DIAGRAM



2) Data Definition Language Details (DDL)

Tables

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```
1  |----- TABLES -----
2
3  CREATE TABLE Roles (
4      role_id SERIAL PRIMARY KEY,
5      role_name TEXT NOT NULL
6  );
7
8  CREATE TABLE Users (
9      user_id SERIAL PRIMARY KEY,
10     username TEXT NOT NULL,
11     password TEXT NOT NULL,
12     full_name TEXT NOT NULL,
13     role_id INT,
14     FOREIGN KEY (role_id)
15         REFERENCES Roles(role_id)
16         ON DELETE SET NULL
17 );
18
19 CREATE TABLE Farms (
20     farm_id SERIAL PRIMARY KEY,
21     farm_name TEXT NOT NULL,
22     location TEXT NOT NULL,
23     size_acre NUMERIC
24 );
```

Untitled Save Primary Active neondb

```
25
26 CREATE TABLE Fields (
27     field_id SERIAL PRIMARY KEY,
28     farm_id INT,
29     field_name TEXT NOT NULL,
30     soil_type TEXT,
31     area NUMERIC,
32     FOREIGN KEY (farm_id)
33         REFERENCES Farms(farm_id)
34         ON DELETE CASCADE
35 );
36
37 CREATE TABLE Crops (
38     crop_id SERIAL PRIMARY KEY,
39     field_id INT,
40     crop_name TEXT NOT NULL,
41     planting_date DATE,
42     status TEXT,
43     FOREIGN KEY (field_id)
44         REFERENCES Fields(field_id)
45         ON DELETE CASCADE
46 );
47
48 CREATE TABLE SensorReadings (
49     reading_id SERIAL PRIMARY KEY,
50     field_id INT,
51     temperature INT,
52     moisture INT,
53     reading_date TIMESTAMP,
54     FOREIGN KEY (field_id)
55         REFERENCES Fields(field_id)
56         ON DELETE CASCADE
57 );
```

Connected (12 queries)

Run Explain Analyze

-  crops
-  farms
-  fields
-  roles
-  sensorreadings
-  users

Values

UntitledSavePrimaryActive▼neondb🕒🔗

```
56  );
57  };
58
59  -- ===== DATA =====
60
61  INSERT INTO Roles (role_name) VALUES
62  ('Admin'),
63  ('Technician'),
64  ('Farmer');
65
66  INSERT INTO Users (username, password, full_name, role_id) VALUES
67  ('admin', '1234', 'System Admin', 1),
68  ('tech1', '1234', 'Hydroponic Technician', 2),
69  ('farmer1', '1234', 'Strawberry Grower', 3);
70
71  INSERT INTO Farms (farm_name, location, size_acre) VALUES
72  ('Hydro Strawberry Farm', 'Nicosia', 12),
73  ('Smart Berry Lab', 'Kyrenia', 8);
74
75  INSERT INTO Fields (farm_id, field_name, soil_type, area) VALUES
76  (1, 'Greenhouse A', 'Hydroponic', 3),
77  (1, 'Greenhouse B', 'Hydroponic', 4),
78  (2, 'Vertical Tower 1', 'Hydroponic', 2);
79
80  INSERT INTO Crops (field_id, crop_name, planting_date, status) VALUES
81  (1, 'Strawberry Albion', '2026-03-01', 'Growing'),
82  (2, 'Strawberry Festival', '2026-02-15', 'Ready'),
83  (3, 'Strawberry San Andreas', '2026-03-10', 'Planted');
84
85  INSERT INTO SensorReadings (field_id, temperature, moisture, reading_date) VALUES
86  (1, 22, 70, '2026-05-06 10:00:00'),
87  (2, 24, 65, '2026-05-06 10:05:00'),
88  (3, 21, 75, '2026-05-06 10:10:00');
```

🔌 Ready to connect

▶ Run

UntitledSavePrimaryActive▼neondb🕒🔗

1 INSERT INTO farms(farm_name) VALUES ('Hydro Strawberry Farm');

✔ Connected (1 query)

▶ Run | Explain | Analyze

⚠ ERROR: Farm name already exists! (SQLSTATE P0001)

Fix with AI 🔗

#	field_id	farm_id	field_name	soil_type	area
1	1	1	Greenhouse A	Hydroponic	3
2	2	1	Greenhouse B	Hydroponic	4
3	3	2	Vertical Tower 1	Hydroponic	2

#	crop_id	field_id	crop_name	planting_date	status
1	1	1	Strawberry Albion	2026-03-01	Growing
2	2	2	Strawberry Festival	2026-02-15	Ready
3	3	3	Strawberry San Andreas	2026-03-10	Planted

#	farm_id	farm_name	location	size_acre
1	1	Hydro Strawberry Farm	Nicosia	12
2	2	Smart Berry Lab	Kyrenia	8

#	role_id	role_name
1	1	Admin
2	2	Technician
3	3	Farmer

#	reading_id	field_id	temperature	moisture	reading_date
1	1	1	22	70	2026-05-06 10:00:00
2	2	2	24	65	2026-05-06 10:05:00
3	3	3	21	75	2026-05-06 10:10:00

#	user_id	username	password	full_name	role_id
1	1	admin	1234	System Admin	1
2	2	tech1	1234	Hydroponic Technician	2
3	3	farmer1	1234	Strawberry Grower	3

```
1 ALTER TABLE fields
2 ADD CONSTRAINT unique_field_name UNIQUE (field_name);
```

```
1 ALTER TABLE farms
2 ADD CONSTRAINT unique_farm_name UNIQUE (farm_name);
```

```
1 ✓ ALTER TABLE crops
2   ADD CONSTRAINT unique_crop_field
3   UNIQUE (crop_name, field_id);
```

```
1 ✓ DELETE FROM farms
2   WHERE farm_id NOT IN (
3       SELECT MIN(farm_id)
4       FROM farms
5       GROUP BY farm_name
6   );|
```

```
1 ✓ DELETE FROM crops
2   WHERE crop_id NOT IN (
3       SELECT MIN(crop_id)
4       FROM crops
5       GROUP BY crop_name, field_id
6   );
```

3) Data Manipulation Language (DML)

1.Query: This query finds fields whose average temperature is higher than the overall average temperature of all sensor readings.

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```
1 SELECT f.field_name,
2     AVG(s.temperature) AS avg_temp
3 FROM sensorreadings s
4 JOIN fields f ON s.field_id = f.field_id
5 GROUP BY f.field_name
6 HAVING AVG(s.temperature) > (
7     SELECT AVG(temperature)
8     FROM sensorreadings
9 );
```

✓ Connected (1 query)

▶ Run

Explain

Analyze

215ms1 row

#	field_name	avg_temp
1	Greenhouse B	24.000000000000000

StatisticsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Highest Moisture Field

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

	field_name	avg_temp
▶	Greenhouse B	24,000000000000...
*		

2.Query: This query calculates the total number of crops in each farm by joining farms, fields, and crops tables.

UntitledSavePrimaryActive▼neondb⌚⚙

```
1 SELECT fa.farm_name, COUNT(c.crop_id) AS total_crops
2 FROM farms fa
3 JOIN fields f ON fa.farm_id = f.farm_id
4 JOIN crops c ON f.field_id = c.field_id
5 GROUP BY fa.farm_name
6 ORDER BY total_crops DESC;
```

✓ Connected (1 query)

▶ Run | Explain | Analyze

230ms 2 rc

#	farm_name	total_crops
1	Smart Berry Lab	2
2	Hydro Strawberry Farm	1

StatisticsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Highest Moisture Field

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

	farm_name	total_crops
▶	Hydro Strawber...	2
	Smart Berry Lab	1
*		

3.Query: This query shows the minimum and maximum temperature recorded in each field.

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```
1 SELECT f.field_name,
2       MAX(s.temperature) AS max_temp,
3       MIN(s.temperature) AS min_temp
4 FROM sensorreadings s
5 JOIN fields f ON s.field_id = f.field_id
6 GROUP BY f.field_name;
```

✓ Connected (1 query)

▶ Run

Explain

Analyze

195ms3 rows

#	field_name	max_temp	min_temp
1	Vertical Tower 1	23	23
2	Greenhouse B	22	22
3	Greenhouse A	22	22

StatisticsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Low Moisture Analysis

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

	field_name	max_temp	min_temp
▶	Vertical Tower 1	23	23
	Greenhouse B	22	22
	Greenhouse A	22	22
*			

4.Query: This query finds the field with the highest moisture value by calculating the maximum moisture per field and returning the top result.

UntitledSavePrimaryActive▼neondb🕒⚙️

```
1 SELECT f.field_name,
2       MAX(s.moisture) AS highest_moisture
3 FROM "fields" f
4 JOIN "sensorreadings" s
5 ON f.field_id = s.field_id
6 GROUP BY f.field_name
7 ORDER BY highest_moisture DESC
8 LIMIT 1;
```

✓ Connected (1 query)

▶ Run

Explain

Analyze

236ms1 row

#	field_name	highest_moisture
1	Greenhouse A	70

StatisticsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Highest Moisture Field

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

	field_name	highest_moisture
▶	Greenhouse A	70
*		

5.Query: This query retrieves the most recent temperature reading for each field by selecting records with the latest reading date from the sensorreadings table.

UntitledSavePrimaryActive▼neondb⌚⚙

```
1 SELECT f.field_name, s.temperature, s.reading_date
2 FROM sensorreadings s
3 JOIN fields f ON s.field_id = f.field_id
4 WHERE s.reading_date = (
5     SELECT MAX(s2.reading_date)
6     FROM sensorreadings s2
7     WHERE s2.field_id = s.field_id
8 );
```

✓ Connected (1 query)

▶ Run

Explain

Analyze

207ms2 rows

#	field_name	temperature	reading_date
1	Greenhouse A	22	2026-05-06 10:00:00
2	Greenhouse B	24	2026-05-06 10:05:00

StatisticsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Highest Moisture Field

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

	field_name	temperature	reading_date
▶	Greenhouse A	22	6.05.2026 10:00
	Greenhouse B	24	6.05.2026 10:05
*			

6.Query: This query lists all fields that do not currently contain any crops.

Untitled

Save

Primary Active

neondb

1

2

3

4

SELECT f.field_name

FROM fields f

LEFT JOIN crops c ON f.field_id = c.field_id

WHERE c.crop_id IS NULL;

Connected (1 query)

Run

Explain

Analyze

246ms

No result

Statement executed successfully

StatisticsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

What you want to see?

Above Avg Temperature

Crops Per Farm

Min Max Temperature

Low Moisture Analysis

Latest Sensor Data

Empty Fields

BACK

STATISTIC TABLE VIEW

	field_name
*	

Update Query: This query updates the status of a specific crop to "Harvested" based on its ID. It simulates a real-world scenario where a crop has completed its growth cycle.

Untitled

Save

Primary

Active

neondb

1

2

3

4

5

6

UPDATE crops

SET status = 'Harvested'

WHERE crop_id = 1;

SELECT crop_id, crop_name, status

FROM crops

WHERE crop_id = 1;

Connected (2 queries)

Run

Explain

Analyze

189ms

1 row

1: UPDATE 1

2: SELECT 1

#	crop_id	crop_name	status
1	1	Strawberry Albion	Harvested

Untitled

Save

Primary

Active

neondb

1

2

3

DELETE FROM sensorreadings

WHERE reading_id = 3;

SELECT * FROM sensorreadings;

Connected (2 queries)

Run

Explain

Analyze

224ms

2 rows

1: DELETE 1

2: SELECT 2

#	reading_id	field_id	temperature	moisture	reading_date
1	1	1	22	70	2026-05-06 10:00:00
2	2	2	24	65	2026-05-06 10:05:00

Delete Query: This query deletes a specific sensor reading from the system using its ID. It is useful for removing incorrect or outdated data.

This SQL view simplifies data access by combining farm, field, and crop data into a single, easy-to-query dashboard.

1

2

3

4

5

6

7

8

9

10

11

12

13

14

```
CREATE OR REPLACE VIEW v_crop_monitoring_dashboard AS
SELECT
  f.farm_id,
  f.farm_name,
  fi.field_id,
  fi.field_name,
  fi.soil_type AS farming_method,
  c.crop_id,
  c.status AS crop_status,
  c.planting_date
FROM farms f
JOIN fields fi ON f.farm_id = fi.farm_id
JOIN crops c ON fi.field_id = c.field_id;
SELECT * FROM v_crop_monitoring_dashboard;
```

Ready to connect

Run

171ms3 rows

#	farm_id	farm_name	field_id	field_name	farming_method	crop_id	crop_status	planting_date
1	2	Smart Berry Lab	3	Vertical Tower 1	Hydroponic	3	Planted	2026-03-10
2	1	Hydro Strawberry Farm	1	Greenhouse A	Hydroponic	1	Healthy	2026-03-01
3	1	Hydro Strawberry Farm	2	Greenhouse B	Hydroponic	2	Healthy	2026-02-15

3) PL/SQL QUERIES

1, This procedure inserts a new crop into the database while preventing duplicate crop entries in the same field.

Untitled

Save

Primary Active

neondb

```
1 CREATE OR REPLACE PROCEDURE add_crop(  
2     p_field_id INT,  
3     p_crop_name TEXT,  
4     p_date DATE  
5 )  
6 LANGUAGE plpgsql  
7 AS $$  
8 BEGIN  
9     IF NOT EXISTS (  
10         SELECT 1 FROM crops  
11         WHERE field_id = p_field_id AND crop_name = p_crop_name  
12     ) THEN  
13         INSERT INTO crops(field_id, crop_name, planting_date, status)  
14         VALUES (p_field_id, p_crop_name, p_date, 'New');  
15     ELSE  
16         RAISE NOTICE 'Crop already exists in this field';  
17     END IF;  
18 END;  
19 $$;  
20 CALL add_crop(1, 'StrawberryX', '2026-05-14');
```

✓ Connected (2 queries)

▶ Run

Explain

Analyze

275ms

No result

1: CREATE

2: CALL

Statement executed successfully

Untitled

Save

Primary Active

neondb

```
1 SELECT crop_name, field_id FROM crops;
```

✓ Connected (1 query)

▶ Run

Explain

Analyze

197ms

1 row

#	crop_name	field_id
1	Strawberry Festival	2

2, This procedure automatically updates a field's crop status in the database to a specific risk level or "Healthy" by evaluating incoming temperature and moisture sensor data.

The screenshot shows a SQL IDE interface with a query editor, a toolbar, and a results pane. The query editor contains a PL/SQL procedure named `check_sensor_status` that takes `p_field_id`, `p_temperature`, and `p_moisture` as parameters. The procedure uses conditional logic to update the status of a crop in the `crops` table based on temperature and moisture levels. It also includes `RAISE NOTICE` statements to provide feedback. The procedure is then called with `check_sensor_status(1, 35, 50);`. The toolbar includes buttons for `Run`, `Explain`, and `Analyze`. The results pane shows the execution status and a message indicating that the statement was executed successfully.

```
1 CREATE OR REPLACE PROCEDURE check_sensor_status(  
2     p_field_id BIGINT,  
3     p_temperature NUMERIC,  
4     p_moisture NUMERIC  
5 )  
6 LANGUAGE plpgsql  
7 AS $$  
8 BEGIN  
9     IF p_temperature > 30 THEN  
10        UPDATE crops  
11        SET status = 'High Temperature Risk'  
12        WHERE field_id = p_field_id;  
13  
14        RAISE NOTICE 'Temperature is too high. Crop status updated.';  
15    ELSEIF p_moisture < 40 THEN  
16        UPDATE crops  
17        SET status = 'Low Moisture Risk'  
18        WHERE field_id = p_field_id;  
19  
20        RAISE NOTICE 'Moisture is too low. Crop status updated.';  
21    ELSE  
22        UPDATE crops  
23        SET status = 'Healthy'  
24        WHERE field_id = p_field_id;  
25  
26        RAISE NOTICE 'Field conditions are normal.';  
27    END IF;  
28 END;  
29 $$;  
30 CALL check_sensor_status(1, 35, 50);
```

Connected (2 queries)

Run Explain Analyze 228ms No result

1: CREATE
Temperature is too high. Crop status updated.
Statement executed successfully

3, This procedure inserts a new account record into the users table using the provided username, password, full name, and role ID.

The screenshot shows a SQL IDE interface with a query editor, a toolbar, and a results pane. The query editor contains a PL/SQL procedure named `add_new_user` that takes `p_username`, `p_password`, `p_full_name`, and `p_role_id` as parameters. The procedure inserts a new record into the `users` table using the provided values. It also includes a `RAISE NOTICE` statement to provide feedback. The procedure is then called with `add_new_user('zehra', '1234', 'Zehra Tas', 1);`. The toolbar includes buttons for `Run`, `Explain`, and `Analyze`. The results pane shows the execution status and a message indicating that the statement was executed successfully.

```
1 CREATE OR REPLACE PROCEDURE add_new_user(  
2     p_username TEXT,  
3     p_password TEXT,  
4     p_full_name TEXT,  
5     p_role_id BIGINT  
6 )  
7 LANGUAGE plpgsql  
8 AS $$  
9 BEGIN  
10    INSERT INTO users(  
11        username,  
12        password,  
13        full_name,  
14        role_id  
15    )  
16    VALUES (  
17        p_username,  
18        p_password,  
19        p_full_name,  
20        p_role_id  
21    );  
22  
23    RAISE NOTICE 'New user added successfully.';  
24 END;  
25 $$;  
26 CALL add_new_user('zehra', '1234', 'Zehra Tas', 1);
```

Connected (2 queries)

Run Explain Analyze 197ms No result

1: CREATE 2: CALL
Statement executed successfully
New user added successfully.

4, This function returns the total number of crops in a given field for analytical purposes.

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```
1 CREATE OR REPLACE FUNCTION get_crop_count(p_field_id INT)
2 RETURNS INT
3 LANGUAGE plpgsql
4 AS $$
5 DECLARE total INT;
6 BEGIN
7     SELECT COUNT(*) INTO total
8     FROM crops
9     WHERE field_id = p_field_id;
10
11     RETURN total;
12 END;
13 $$;
14 SELECT get_crop_count(1);
```

 Connected (2 queries)

 Run | Explain Analyze 205ms 1 row

1: CREATE 2: SELECT 1

#	get_crop_count
1	0

5, TRIGGER Detects low moisture levels and warns the system to prevent crop damage.

Untitled Save Primary Active neondb  

```
1 CREATE OR REPLACE FUNCTION low_moisture_warning()
2 RETURNS TRIGGER AS $$
3 BEGIN
4     IF NEW.moisture < 60 THEN
5         RAISE NOTICE 'Warning: Low moisture detected!';
6     END IF;
7
8     RETURN NEW;
9 END;
10 $$ LANGUAGE plpgsql;
11 CREATE TRIGGER trg_low_moisture
12 AFTER INSERT ON sensorreadings
13 FOR EACH ROW
14 EXECUTE FUNCTION low_moisture_warning();
```

 Connected (2 queries)

 Run | Explain Analyze 201ms No result

1: CREATE 2: CREATE

Statement executed successfully

SQL Editor

production

Saved History

insert new sensor reading
May 16, 2026 - 4:51pm

insert sensor reading for field 2
May 16, 2026 - 4:51pm

trigger for low moisture warning on sensor readings
May 16, 2026 - 4:50pm

low moisture function
May 16, 2026 - 4:50pm

Untitled Save Primary Active neondb

```

1 INSERT INTO sensorreadings(field_id, temperature, moisture, reading_date)
2 VALUES (2, 25, 40, NOW());

```

Connected (1 query)

Run Explain Analyze

Warning: Low moisture detected!

6, This trigger prevents insertion of farms with duplicate names to maintain data integrity.

Untitled Save Primary Active neondb

```

1 CREATE OR REPLACE FUNCTION prevent_duplicate_farm()
2 RETURNS TRIGGER
3 LANGUAGE plpgsql
4 AS $$
5 BEGIN
6     IF EXISTS (
7         SELECT 1 FROM farms WHERE farm_name = NEW.farm_name
8     ) THEN
9         RAISE EXCEPTION 'Farm name already exists!';
10    END IF;
11
12    RETURN NEW;
13 END;
14 $$;
15
16 CREATE TRIGGER trg_unique_farm
17 BEFORE INSERT ON farms
18 FOR EACH ROW
19 EXECUTE FUNCTION prevent_duplicate_farm();

```

Connected (2 queries)

Run Explain Analyze 190ms No result

1: CREATE 2: CREATE

Statement executed successfully

7, This trigger logs a "Critical temperature detected!" message into an alerts table whenever a new sensor reading exceeds 30 degrees.

Untitled

Save

Primary Active

neondb

```
1 CREATE TABLE alerts (  
2   alert_id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
3   field_id BIGINT,  
4   message TEXT,  
5   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
6 );  
7 CREATE OR REPLACE FUNCTION high_temperature_alert()  
8 RETURNS TRIGGER AS $$  
9 BEGIN  
10  IF NEW.temperature > 30 THEN  
11    INSERT INTO alerts(field_id, message)  
12    VALUES (  
13      NEW.field_id,  
14      'Critical temperature detected!'  
15    );  
16  END IF;  
17  RETURN NEW;  
18 END;  
19 $$ LANGUAGE plpgsql;  
20 CREATE TRIGGER trg_high_temperature  
21 AFTER INSERT ON sensorreadings  
22 FOR EACH ROW  
23 EXECUTE FUNCTION high_temperature_alert();  
24
```

Connected (3 queries)

Run

Explain

Analyze

161ms No result

1: CREATE 2: CREATE 3: CREATE

Statement executed successfully

8, This PL/pgSQL function iterates through every field in the database, calculates their average temperature and moisture levels from recent sensor readings, and executes a status check procedure to evaluate and update their crop conditions.

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```
1 CREATE OR REPLACE FUNCTION monitor_all_fields()
2 RETURNS VOID AS $$
3 DECLARE
4     field_record RECORD;
5     avg_temp NUMERIC;
6     avg_moisture NUMERIC;
7 BEGIN
8     FOR field_record IN SELECT field_id FROM fields LOOP
9
10        SELECT AVG(temperature), AVG(moisture)
11        INTO avg_temp, avg_moisture
12        FROM SensorReadings
13        WHERE field_id = field_record.field_id;
14
15        IF avg_temp IS NOT NULL THEN
16            CALL check_sensor_status(field_record.field_id, avg_temp, avg_moisture);
17        END IF;
18    END LOOP;
19 END;
20 $$ LANGUAGE plpgsql;
21 SELECT monitor_all_fields();
```

Connected (2 queries)

▶ Run

Explain

Analyze

219ms1 row

1: CREATE2: SELECT 1

#	monitor_all_fields
1	

⚠ Field conditions are normal.

⚠ Field conditions are normal.

⌂⬇🖨

4)DB Connection

This section explains how the application connects to the PostgreSQL database using **Npgsql**.

The connection is established using a connection string obtained from the cloud database provider.

A simple SQL query is executed to verify user login credentials (username and password).

```
24 private void btnLogin_Click(object sender, EventArgs e)
25 {
26     using (var conn = new NpgsqlConnection(connString))
27     {
28         conn.Open();
29
30         string query = @"SELECT * FROM Users
31             WHERE username=@u AND password=@p";
```

```
9 namespace SmartFarmingSystem
10 {
11     E Başvuru
12     public partial class LoginForm : Form
13     {
14         string connString = "Host=ep-bold-surf-apre6yz3.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npq_nqxUsDFfP10g;SSL Mode=Require;Trust Server Certificate=true;";
```

Connect your app manually

Copy and paste your production branch's connection details into your app's .env file

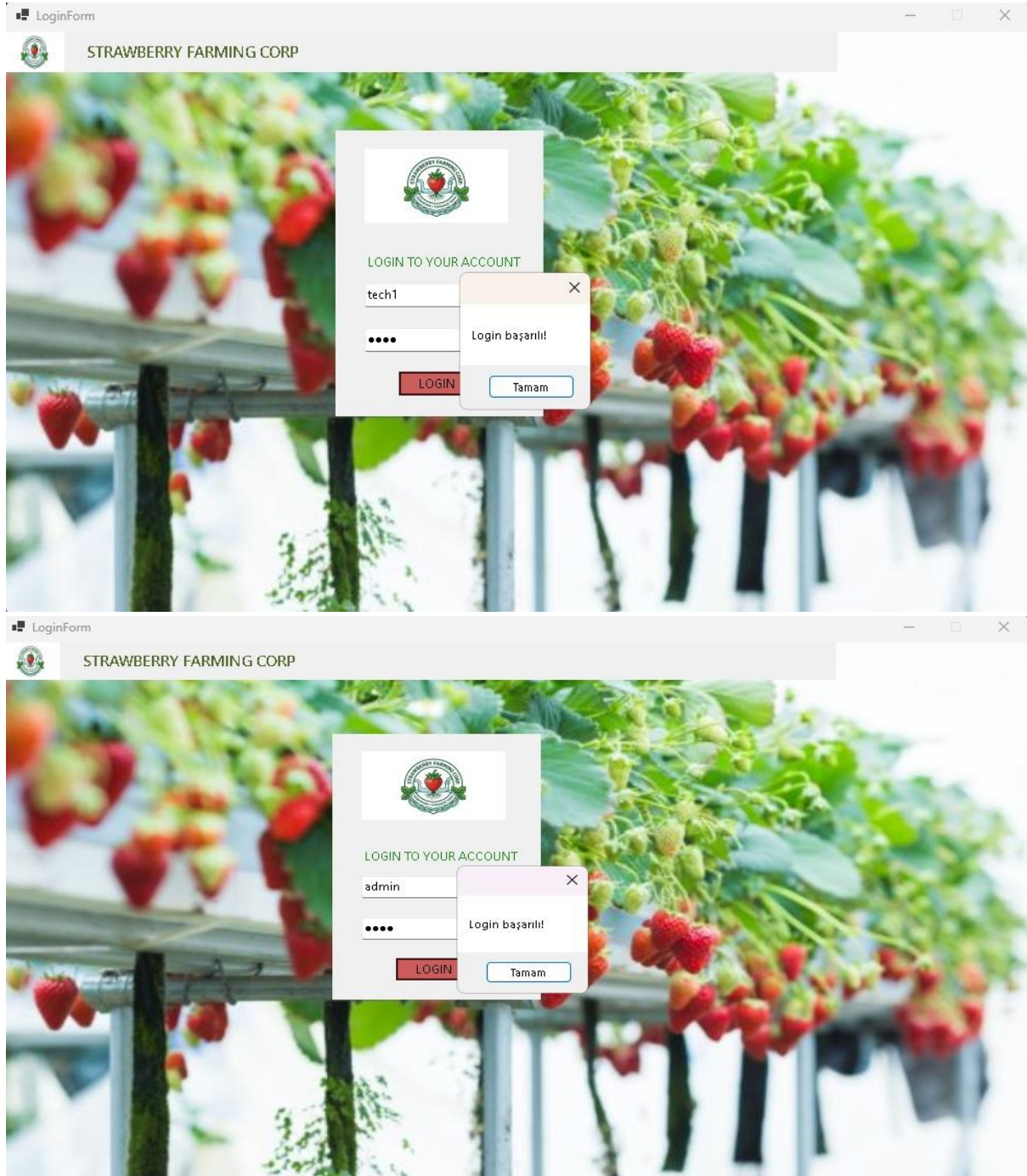
Connection string

Connection parameters

Host	ep-bold-surf-apre6yz3.c-7.us-east-1.aws.neon.tech
Database	neondb
Role	neondb_owner
Password	*****
Pooler host	ep-bold-surf-apre6yz3-pooler.c-7.us-east-1.aws.neon.tech

5)Codes and GUI

Login Page



```
SmartFarmingSystem
SmartFarmingSystem.LoginForm
lblPassword_Click(object sender, EventArgs e)

1
2 using System;
3 using System.Collections.Generic;
4 using System.ComponentModel;
5 using System.Data;
6 using System.Drawing;
7 using System.Text;
8 using System.Windows.Forms;
9 using Npgsql;
10 namespace SmartFarmingSystem
11 {
12     6 bayrak
13     public partial class LoginForm : Form
14     {
15         string connString = "Host=ep-bold-surf-apr6yz3.c-7.us-east-1.amazonaws.com;Database=neondb;Username=neondb_owner;Password=npqUsDFFP1Bg;SSL Mode=Require;Trust Server Certificate=true;";
16         public LoginForm()
17         {
18             InitializeComponent();
19             this.StartPosition = FormStartPosition.CenterScreen;
20             this.Size = new Size(1000, 600);
21             this.FormBorderStyle = FormBorderStyle.FixedSingle;
22             this.MaximizeBox = false;
23         }
24         1 bayrak
25         private void btnLogin_Click(object sender, EventArgs e)
26         {
27             using (var conn = new NpgsqlConnection(connString))
28             {
29                 conn.Open();
30                 string query = @"SELECT role_id FROM Users
31                               WHERE username=@u AND password=@p";
32                 using (var cmd = new NpgsqlCommand(query, conn))
33                 {
34                     cmd.Parameters.AddWithValue("@u", txtUsername.Text);
35                     cmd.Parameters.AddWithValue("@p", txtPassword.Text);
36                     var result = cmd.ExecuteScalar();
37                     if (result != null)
38                     {
39                         int roleId = Convert.ToInt32(result);
40                         MessageBox.Show("Login başarılı!");
41                     }
42                 }
43             }
44         }
45     }
46     41
47     {
48         int roleId = Convert.ToInt32(result);
49         MessageBox.Show("Login başarılı!");
50         Dashboard d = new Dashboard(roleId); // ROLE GÖNDERİYORUZ
51         d.Show();
52         this.Hide();
53     }
54     else
55     {
56         MessageBox.Show("Hatalı giriş!");
57     }
58 }
59 }
```


Dashboard when farmer logs in

Dashboard

STRAWBERRY FARMING CORP

Welcome! Have a nice day

WHAT YOU WANT TO DO?

Manage Crops

Manage Farms

Manage Sensors

Manage Field

Statistics

Log Out

Crops: 3

Farms: 2

	field_name	temperature	moisture	reading_date
▶	Greenhouse A	22	70	6.05.2026 10:00
	Greenhouse B	22	70	6.05.2026 10:00
	Vertical Tower 1	23	54	14.05.2026 16:09
*				

Dashboard when admin logs in

Dashboard

STRAWBERRY FARMING CORP

Welcome! Have a nice day

WHAT YOU WANT TO DO?

Manage Crops

Manage Farms

Manage Sensors

Manage Field

Statistics

Manage Users

Log Out

Crops: 3

Farms: 2

	field_name	temperature	moisture	reading_date
▶	Greenhouse A	22	70	6.05.2026 10:00
	Greenhouse B	24	65	6.05.2026 10:05
*				

```
Dashboard.cs | ManageUsers.cs | FieldForm.cs [Tasarım] | ManageUsers.cs [Tasarım] | Dashboard.cs [Tasarım] | LoginForm.cs | LoginForm.cs [Tasarım] | StatisticForm.cs
SmartFarmingSystem
1 /stem;
2 /stem.Collections.Generic;
3 /stem.ComponentModel;
4 /stem.Data;
5 /stem.Drawing;
6 /stem.Text;
7 /stem.Windows.Forms;
8 Npgsql;
9
10 namespace SmartFarmingSystem
11 {
12     using System.Data;
13     using System.Data.SqlClient;
14     public partial class Dashboard : Form
15     {
16         string connString = "Host=ep-bold-surf-apr6y23.c-7.us-east-1.amazonaws.com;Database=neondb;Username=neondb_owner;Password=npq_nqxUsDFP10g;SSL Mode=Require;Trust Server Certificate=true;";
17         int userRole;
18
19         public Dashboard(int role)
20         {
21             InitializeComponent();
22             userRole = role;
23         }
24
25         public Dashboard()
26         {
27             InitializeComponent();
28             this.StartPosition = FormStartPosition.CenterScreen;
29             this.Size = new Size(1000, 600);
30             this.FormBorderStyle = FormBorderStyle.FixedSingle;
31             this.MaximizeBox = false;
32         }
33
34         private void Dashboard_Load(object sender, EventArgs e)
35         {
36             LoadDashboardData();
37             if (userRole != 1)
38             {
39                 btnManageUsers.Visible = false;
40             }
41         }
42
43         void LoadDashboardData()
44         {
45             using (var conn = new NpgsqlConnection(connString))
46             {
47                 conn.Open();
48
49                 // toplam crops
50                 using (var cmd = new NpgsqlCommand("SELECT COUNT(*) FROM Crops", conn))
51                 {
52                     lblCrops.Text = cmd.ExecuteScalar().ToString();
53                 }
54
55                 // toplam farms
56                 using (var cmd = new NpgsqlCommand("SELECT COUNT(*) FROM Farms", conn))
57                 {
58                     lblFarms.Text = cmd.ExecuteScalar().ToString();
59                 }
60
61                 // sensör tablosu
62                 string query = @"SELECT f.field_name, s.temperature, s.moisture, s.reading_date
63                               FROM SensorReadings s
64                               JOIN Fields f ON s.field_id = f.field_id";
65
66                 using (var da = new NpgsqlDataAdapter(query, conn))
67                 {
68                     DataTable dt = new DataTable();
69                     da.Fill(dt);
70                     dataGridView1.DataSource = dt;
71                 }
72             }
73         }
74
75         private void lbl3_Click(object sender, EventArgs e)
76         {
77         }
78
79         private void panel2_Paint(object sender, PaintEventArgs e)
80         {
81         }
82
83         private void button2_Click(object sender, EventArgs e)
84         {
85             FarmsForm f = new FarmsForm(); // şimdilik bunu aç
86             f.Show();
87         }
88     }
89 }
```

```
Dashboard.cs | ManageUsers.cs | FieldForm.cs [Tasarım] | ManageUsers.cs [Tasarım] | Dashboard.cs [Tasarım] | LoginForm.cs | LoginForm.cs [Tasarım] | StatisticForm.cs
SmartFarmingSystem
43 void LoadDashboardData()
44 {
45     using (var conn = new NpgsqlConnection(connString))
46     {
47         conn.Open();
48
49         // toplam crops
50         using (var cmd = new NpgsqlCommand("SELECT COUNT(*) FROM Crops", conn))
51         {
52             lblCrops.Text = cmd.ExecuteScalar().ToString();
53         }
54
55         // toplam farms
56         using (var cmd = new NpgsqlCommand("SELECT COUNT(*) FROM Farms", conn))
57         {
58             lblFarms.Text = cmd.ExecuteScalar().ToString();
59         }
60
61         // sensör tablosu
62         string query = @"SELECT f.field_name, s.temperature, s.moisture, s.reading_date
63                       FROM SensorReadings s
64                       JOIN Fields f ON s.field_id = f.field_id";
65
66         using (var da = new NpgsqlDataAdapter(query, conn))
67         {
68             DataTable dt = new DataTable();
69             da.Fill(dt);
70             dataGridView1.DataSource = dt;
71         }
72     }
73 }
74
75 private void lbl3_Click(object sender, EventArgs e)
76 {
77 }
78
79 private void panel2_Paint(object sender, PaintEventArgs e)
80 {
81 }
82
83 private void button2_Click(object sender, EventArgs e)
84 {
85     FarmsForm f = new FarmsForm(); // şimdilik bunu aç
86     f.Show();
87 }
88 }
```

```
Dashboard.cs x ManageUsers.cs FieldsForm.cs [Tasann] ManageUsers.cs [Tasann] Dashboard.cs [Tasann] LoginForm.cs LoginForm.cs [Tasann] StatisticsForm.cs
SmartFarmingSystem SmartFarmingSystem.Dashboard btnManageUsers_Click(object sender, EventArgs e)
85 private void button2_Click(object sender, EventArgs e)
86 {
87     FarmsForm f = new FarmsForm(); // şimdilik bunu aç
88     f.Show();
89     this.Close();
90 }
91
92 1 bagyuru
93 private void button4_Click(object sender, EventArgs e)
94 {
95     FieldsForm f = new FieldsForm();
96     f.Show();
97     this.Close();
98 }
99
100 0 bagyuru
101 private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
102 {
103 }
104
105 1 bagyuru
106 private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
107 {
108 }
109
110 1 bagyuru
111 private void buttonlogout_Click(object sender, EventArgs e)
112 {
113     LoginForm login = new LoginForm();
114     login.Show();
115     this.Close();
116 }
117
118 1 bagyuru
119 private void btnCrops_Click(object sender, EventArgs e)
120 {
121     CropsForm form = new CropsForm();
122     form.Show();
123     this.Close();
124 }
125
126 1 bagyuru
127 private void btnSensors_Click(object sender, EventArgs e)
128 {
129     SensorReadingsForm f = new SensorReadingsForm();
130     f.Show();
131     this.Close();
132 }
133
134 100% 0 2 Sat: 146, Kiri: 25 BSL CRLF UTF-8 with BOM
```

```
Dashboard.cs x ManageUsers.cs FieldsForm.cs [Tasann] ManageUsers.cs [Tasann] Dashboard.cs [Tasann] LoginForm.cs LoginForm.cs [Tasann] StatisticsForm.cs
SmartFarmingSystem SmartFarmingSystem.Dashboard btnManageUsers_Click(object sender, EventArgs e)
130 1 bagyuru
131 private void btnStatistics_Click(object sender, EventArgs e)
132 {
133     StatisticsForm s = new StatisticsForm();
134     s.Show();
135     this.Close();
136 }
137
138 1 bagyuru
139 private void label2_Click(object sender, EventArgs e)
140 {
141 }
142
143 1 bagyuru
144 private void btnManageUsers_Click(object sender, EventArgs e)
145 {
146     ManageUsers frm = new ManageUsers(this);
147     frm.Show();
148     this.Hide();
149 }
150
```

Users Managemnet (for admins only)

Adding user

ManageUsers

STRAWBERRY FARMING CORP

Welcome! Have a nice day

 USERS MANAGEMENT

User Name:

Password:

Full Name:

Role:

ADD

UPDATE

DELETE

BACK

	user_id	username	full_name	role_name
	1	admin	System Admin	Admin
	2	tech1	Hydroponic Te...	Technician
	3	farmer1	Strawberry Gro...	Farmer
▶	7	Emre	Emre Özel	Farmer
*				

Updating user

ManageUsers

STRAWBERRY FARMING CORP

Welcome! Have a nice day

 USERS MANAGEMENT

User Name:

Password:

Full Name:

Role:

ADD

UPDATE

DELETE

BACK

	user_id	username	full_name	role_name
	1	admin	System Admin	Admin
	2	tech1	Hydroponic Te...	Technician
	3	farmer1	Strawberry Gro...	Farmer
▶	7	Emre	Emre Özel	Admin
*				

Deleting User

ManageUsers

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

 USERS MANAGEMENT

User Name:

Password:

Full Name:

Role:

	user_id	username	full_name	role_name
▶	1	admin	System Admin	Admin
	2	tech1	Hydroponic Te...	Technician
	3	farmer1	Strawberry Gro...	Farmer
•				

```
ManageUsers.cs [Tasarruf] | FieldForm.cs [Tasarruf] | ManageUsers.cs [Tasarruf] | Dashboard.cs [Tasarruf] | LoginForm.cs | LoginForm.cs [Tasarruf] | StatisticForm.cs |
SmartFarmingSystem | SmartFarmingSystem.ManageUsers | btnAdd_Click(object sender, EventArgs e)
1 using Npgsql;
2 using System;
3 using System.Collections.Generic;
4 using System.ComponentModel;
5 using System.Data;
6 using System.Drawing;
7 using System.Text;
8 using System.Windows.Forms;
9
10 namespace SmartFarmingSystem
11 {
12     5 bayram
13     public partial class ManageUsers : Form
14     {
15         string connString = "Host=ep-bold-surf-apre6yz3.c-7.us-east-1.amazonaws.com;Database=neondb;Username=neondb_owner;Password=npq_nqxUsDffP18g;SSL Mode=Require;Trust Server Certificate=true;";
16         Dashboard dashboard;
17         4 bayram
18         void LoadUsers()
19         {
20             using (var conn = new NpgsqlConnection(connString))
21             {
22                 conn.Open();
23
24                 string q = @"SELECT u.user_id, u.username, u.full_name, r.role_name
25 FROM **users** u
26 LEFT JOIN **roles** r ON u.role_id = r.role_id";
27
28                 using (var da = new NpgsqlDataAdapter(q, conn))
29                 {
30                     DataTable dt = new DataTable();
31                     da.Fill(dt);
32                     dataGridView1.DataSource = dt;
33                 }
34             }
35         }
36         1 bayram
37         void LoadRoles()
38         {
39             using (var conn = new NpgsqlConnection(connString))
40             {
41                 conn.Open();
42
43                 string q = "SELECT role_id, role_name FROM roles";
44
45                 using (var da = new NpgsqlDataAdapter(q, conn))
46                 {
47                     DataTable dt = new DataTable();
48                     da.Fill(dt);
49                     cmbRole.DataSource = dt;
50                 }
51             }
52         }
53     }
54 }
```

```
ManageUsers.cs | FieldForm.cs [Tasnm] | ManageUsers.cs [Tasnm] | Dashboard.cs [Tasnm] | LoginForm.cs | LoginForm.cs [Tasnm] | StatisticForm.cs
SmartFarmingSystem | SmartFarmingSystem.ManageUsers | btnAdd_Click(object sender, EventArgs e)
44:         DataTable dt = new DataTable();
45:         da.Fill(dt);
46:
47:         cmbRole.DataSource = dt;
48:         cmbRole.DisplayMember = "role_name";
49:         cmbRole.ValueMember = "role_id";
50:     }
51: }
52:
53:
54: 1 bagpuru
55: public ManageUsers(Dashboard d)
56: {
57:     InitializeComponent();
58:     dashboard = d;
59: }
60:
61: 0 bagpuru
62: private void panelTop_Paint(object sender, PaintEventArgs e)
63: {
64: }
65:
66: 1 bagpuru
67: private void ManageUsers_Load(object sender, EventArgs e)
68: {
69:     LoadUsers();
70:     LoadRoles();
71: }
72:
73: 1 bagpuru
74: private void btnAdd_Click(object sender, EventArgs e)
75: {
76:     using (var conn = new NpgsqlConnection(connString))
77:     {
78:         conn.Open();
79:
80:         string q = @"INSERT INTO users(username, password, full_name, role_id)
81:             VALUES(@u, @p, @f, @r)";
82:
83:         using (var cmd = new NpgsqlCommand(q, conn))
84:         {
85:             cmd.Parameters.AddWithValue("@u", txtUsername.Text);
86:             cmd.Parameters.AddWithValue("@p", txtPassword.Text);
87:             cmd.Parameters.AddWithValue("@f", txtFullName.Text);
88:             cmd.Parameters.AddWithValue("@r", cmbRole.SelectedValue);
89:
90:             cmd.ExecuteNonQuery();
91:         }
92:     }
93: }
```

```
ManageUsers.cs | FieldForm.cs [Tasnm] | ManageUsers.cs [Tasnm] | Dashboard.cs [Tasnm] | LoginForm.cs | LoginForm.cs [Tasnm] | StatisticForm.cs
SmartFarmingSystem | SmartFarmingSystem.ManageUsers | btnAdd_Click(object sender, EventArgs e)
89:     }
90:     LoadUsers();
91: }
92:
93: int selectedUserId;
94:
95: 0 bagpuru
96: private void dataGridView1_CellClick(object sender, DataGridViewCellEventArgs e)
97: {
98:     selectedUserId = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
99:
100:     MessageBox.Show("ID: " + selectedUserId);
101:
102:     txtUsername.Text = dataGridView1.CurrentRow.Cells["username"].Value.ToString();
103:     txtFullName.Text = dataGridView1.CurrentRow.Cells["full_name"].Value.ToString();
104: }
105:
106: 1 bagpuru
107: private void dataGridView1_SelectionChanged(object sender, EventArgs e)
108: {
109:     if (dataGridView1.CurrentRow != null)
110:     {
111:         selectedUserId = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
112:
113:         txtUsername.Text = dataGridView1.CurrentRow.Cells[1].Value?.ToString();
114:         txtFullName.Text = dataGridView1.CurrentRow.Cells[2].Value?.ToString();
115:     }
116: }
117:
118: 1 bagpuru
119: private void btnUpdate_Click(object sender, EventArgs e)
120: {
121:
122:     using (var conn = new NpgsqlConnection(connString))
123:     {
124:         conn.Open();
125:
126:         string q = @"UPDATE **users**
127:             SET username=@u, password=@p, full_name=@f, role_id=@r
128:             WHERE user_id=@id";
129:
130:         using (var cmd = new NpgsqlCommand(q, conn))
131:         {
132:             cmd.Parameters.AddWithValue("@u", txtUsername.Text);
133:             cmd.Parameters.AddWithValue("@p", txtPassword.Text);
134:             cmd.Parameters.AddWithValue("@f", txtFullName.Text);
135:             cmd.Parameters.AddWithValue("@r", cmbRole.SelectedValue);
136:             cmd.Parameters.AddWithValue("@id", selectedUserId);
137:
138:             cmd.ExecuteNonQuery();
139:         }
140:     }
141: }
```

```
ManageUsers.cs | FieldForm.cs [Tasarım] | ManageUsers.cs [Tasarım] | Dashboard.cs [Tasarım] | LoginForm.cs | LoginForm.cs [Tasarım] | StatisticForm.cs
SmartFarmingSystem | SmartFarmingSystem.ManageUsers | btnAdd_Click(object sender, EventArgs e)
137         LoadUsers();
138     }
139
140     1 bayyuru
141     private void btnDelete_Click(object sender, EventArgs e)
142     {
143         if (selectedUserId == 0)
144         {
145             MessageBox.Show("Lütfen kullanıcı seçi!");
146             return;
147         }
148
149         DialogResult r = MessageBox.Show("Silmek istiyor musun?", "Confire", MessageBoxButtons.YesNo);
150
151         if (r == DialogResult.Yes)
152         {
153             using (var conn = new NpgsqlConnection(connString))
154             {
155                 conn.Open();
156
157                 string q = "DELETE FROM \users\ WHERE user_id=@id";
158
159                 using (var cmd = new NpgsqlCommand(q, conn))
160                 {
161                     cmd.Parameters.AddWithValue("@id", selectedUserId);
162                     cmd.ExecuteNonQuery();
163                 }
164             }
165
166             LoadUsers();
167         }
168     }
169
170     1 bayyuru
171     private void btnBack_Click(object sender, EventArgs e)
172     {
173         Dashboard d = new Dashboard(1); // geçici (role da gönderebilirsiniz)
174         d.Show();
175         this.Close();
176     }
177
178     1 bayyuru
179     private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
180     {
181     }
182
183 }
```

100% | Sat: 83, Kiri: 73 | BGL | CRLF | UTF-8 with BOM

Crops management

Adding crop

CropsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

CROPS MANAGEMENT

Crop Name: Strawberry test

Field: Greenhouse A

Date: 14 Mayıs 2026 Perşembe

ADDUPDATEDELETE

BACK

	crop_id	crop_name	field_name	planting_date	status
▶	2	Strawberry Festi...	Greenhouse B	15.02.2026	Ready
	3	Strawberry San ...	Vertical Tower 1	10.03.2026	Planted
	1	Strawberry Albi...	Greenhouse A	1.03.2026	Harvested
	18	Strawberry test	Greenhouse A	14.05.2026	New
*					

Updating crop

CropsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

CROPS MANAGEMENT

Crop Name: Strawberry test

Field: Greenhouse B

Date: 14 Mayıs 2026 Perşembe

ADDUPDATEDELETE

BACK

	crop_id	crop_name	field_name	planting_date	status
▶	2	Strawberry Festi...	Greenhouse B	15.02.2026	Ready
	3	Strawberry San ...	Vertical Tower 1	10.03.2026	Planted
	1	Strawberry Albi...	Greenhouse A	1.03.2026	Harvested
	18	Strawberry test	Greenhouse B	14.05.2026	Updated
*					

Deleting crop

The screenshot displays the 'CropsForm' application window. The title bar reads 'CropsForm'. The main window has a header with the 'STRAWBERRY FARMING CORP' logo and the text 'Welcome! Have a nice day'. Below the header, the 'CROPS MANAGEMENT' section contains a form with the following fields:

- Crop Name: Strawberry test
- Field: Greenhouse B
- Date: 14 Mayıs 2026 Perşembe

Below the form are buttons for 'ADD', 'UPDATE', 'DELETE', and 'BACK'.

To the right of the form is a table with the following data:

	crop_id	crop_name	field_name	planting_date	status
▶	2	Strawberry Festi...	Greenhouse B	15.02.2026	Ready
	3	Strawberry San ...	Vertical Tower 1	10.03.2026	Planted
	1	Strawberry Albi...	Greenhouse A	1.03.2026	Harvested
*					

The background shows the C# code for the 'CropsForm' class, which is a partial class for the 'Form' type. The code includes the following methods:

- `LoadCrops()`: A method that connects to a database, executes a query to retrieve crop data, and populates a `DataGridView`.
- `LoadFields()`: A method that clears the `cmbField` dropdown and adds 'Greenhouse A' as an option.

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Data;
5 using System.Drawing;
6 using System.Linq;
7 using System.Windows.Forms;
8 using Npgsql;
9
10 namespace SmartFarmingSystem
11 {
12     public partial class CropsForm : Form
13     {
14         string connString = "Host=ep-bold-surf-apre6yz3.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npg_nqxUsDFP10g;SSL Mode=Require;Trust Server Certificate=true;";
15         int selectedCropId = -1;
16
17         public CropsForm()
18         {
19             InitializeComponent();
20             this.Load += CropsForm_Load;
21             this.StartPosition = FormStartPosition.CenterScreen;
22             this.Size = new Size(1000, 600);
23             this.FormBorderStyle = FormBorderStyle.FixedSingle;
24             this.MaximizeBox = false;
25         }
26
27         void LoadCrops()
28         {
29             using (var conn = new NpgsqlConnection(connString))
30             {
31                 conn.Open();
32
33                 string query = @"SELECT c.crop_id, c.crop_name, f.field_name, c.planting_date, c.status
34                                FROM crops c
35                                JOIN fields f ON c.field_id = f.field_id";
36
37                 using (var da = new NpgsqlDataAdapter(query, conn))
38                 {
39                     DataTable dt = new DataTable();
40                     da.Fill(dt);
41                     dataGridView1.DataSource = dt;
42                 }
43             }
44         }
45
46         void LoadFields()
47         {
48             cmbField.Items.Clear();
49             cmbField.Items.Add("Greenhouse A");
50         }
51     }
52 }
```

```

46         cmbField.Items.Add("Greenhouse A");
47         cmbField.Items.Add("Greenhouse B");
48         cmbField.Items.Add("Vertical Tower 1");
49     }
50 }
51
52 private void lbl6_Click(object sender, EventArgs e)
53 {
54 }
55
56
57 private void btnAdd_Click(object sender, EventArgs e)
58 {
59     if (txtCropName1.Text == "" || cmbField.Text == "")
60     {
61         MessageBox.Show("Boş alan bırakma!");
62         return;
63     }
64     using (var conn = new NpgsqlConnection(connString))
65     {
66         conn.Open();
67
68         string query = @"INSERT INTO crops (field_id, crop_name, planting_date, status)
69 VALUES (
70     (SELECT field_id FROM fields WHERE field_name=@f LIMIT 1),
71     @name, @date, @status
72 )";
73
74         using (var cmd = new NpgsqlCommand(query, conn))
75         {
76             cmd.Parameters.AddWithValue("@f", cmbField.Text);
77             cmd.Parameters.AddWithValue("@name", txtCropName1.Text);
78             cmd.Parameters.AddWithValue("@date", dtPlantingDate.Value);
79             cmd.Parameters.AddWithValue("@status", "New");
80
81             cmd.ExecuteNonQuery();
82         }
83     }
84     LoadCrops();
85 }
86
87
88 private void CropsForm_Load(object sender, EventArgs e)
89 {
90     LoadCrops();
91     LoadFields();
92 }

```

```

97
98 private void dataGridView1_CellClick(object sender, DataGridViewCellEventArgs e)
99 {
100     if (e.RowIndex >= 0)
101     {
102         var row = dataGridView1.Rows[e.RowIndex];
103
104         // ID
105         if (row.Cells[0].Value != DBNull.Value)
106             selectedCropId = Convert.ToInt32(row.Cells[0].Value);
107
108         // Text alanları
109         txtCropName1.Text = row.Cells[1].Value?.ToString();
110         cmbField.Text = row.Cells[2].Value?.ToString();
111
112         // Date fix (EN KÜTÜK WİSTİM)
113         var v = row.Cells[3].Value;
114
115         if (v != DBNull.Value)
116         {
117             if (v is DateTime dt)
118                 dtPlantingDate.Value = dt;
119
120             else if (v is DateOnly d)
121                 dtPlantingDate.Value = d.ToDateTime(TimeOnly.MinValue);
122         }
123     }
124 }
125
126 private void btnUpdate_Click(object sender, EventArgs e)
127 {
128     if (selectedCropId == -1)
129     {
130         MessageBox.Show("Önce listeden bir kayıt seç!");
131         return;
132     }
133
134     using (var conn = new NpgsqlConnection(connString))
135     {
136         conn.Open();
137
138         string query = @"UPDATE crops
139 SET crop_name=@name,
140     planting_date=@date,
141     field_id = (SELECT field_id FROM fields WHERE field_name=@f LIMIT 1),
142     status=@status
143 WHERE crop_id=@id";
144

```

```
CropForm.cs  X  FieldForm.cs
SmartFarmingSystem  SmartFarmingSystem.CropForm  connString

144         where crop_id=@id";
145
146         using (var cmd = new NpgsqlCommand(query, conn))
147         {
148             cmd.Parameters.AddWithValue("@name", txtCropName1.Text);
149             cmd.Parameters.AddWithValue("@date", dtPlantingDate.Value);
150             cmd.Parameters.AddWithValue("@f", cmbField.Text);
151             cmd.Parameters.AddWithValue("@status", "updated");
152             cmd.Parameters.AddWithValue("@id", selectedCropId);
153
154             int affected = cmd.ExecuteNonQuery();
155
156             if (affected == 0)
157                 MessageBox.Show("Güncellencek kayıt bulunamadı!!");
158         }
159
160         selectedCropId = -1; // reset
161         LoadCrops();
162     }
163
164
165     1 bayguru
166     private void btnDelete_Click(object sender, EventArgs e)
167     {
168         if (dataGridView1.CurrentRow != null)
169         {
170             int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
171
172             using (var conn = new NpgsqlConnection(connString))
173             {
174                 conn.Open();
175
176                 string query = "DELETE FROM crops WHERE crop_id=@id";
177
178                 using (var cmd = new NpgsqlCommand(query, conn))
179                 {
180                     cmd.Parameters.AddWithValue("@id", id);
181                     cmd.ExecuteNonQuery();
182                 }
183             }
184
185             LoadCrops();
186         }
187     }
188
189     1 bayguru
190     private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
191     {
192     }
193
194     1 bayguru
195     private void btnBack_Click(object sender, EventArgs e)
196     {
197         Dashboard d = new Dashboard();
198         d.Show();
199         this.Close();
200     }
```

Farms management

Adding farm

FarmsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

FARM MANAGEMENT

Farm: Hydro test

Location: ciu

Size: 15

ADDUPDATEDELETE

BACK

	farm_id	farm_name	location	size_acre
▶	1	Hydro Strawber...	Nicosia	12
	2	Smart Berry Lab	Kýrenia	10
	10	Hydro test	ciu	15
*				

Updating farm

FarmsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

FARM MANAGEMENT

Farm: Hydro test2

Location: ciu st

Size: 19

ADDUPDATEDELETE

BACK

	farm_id	farm_name	location	size_acre
▶	1	Hydro Strawber...	Nicosia	12
	2	Smart Berry Lab	Kýrenia	10
	10	Hydro test2	ciu st	19
*				

Deleting farm

FarmsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

 FARM MANAGEMENT

Farm:

Location:

Size:

ADD

UPDATE

DELETE

	farm_id	farm_name	location	size_acre
▶	1	Hydro Strawber...	Nicosia	12
	2	Smart Berry Lab	Kyrenia	10
*				

BACK

```
FarmsForm.cs X
SmartFarmingSystem SmartFarmingSystem.FarmsForm connString
1 using Npgsql;
2 using System;
3 using System.Collections.Generic;
4 using System.ComponentModel;
5 using System.Data;
6 using System.Drawing;
7 using System.Text;
8 using System.Windows.Forms;
9 namespace SmartFarmingSystem
10 {
11     5 baguruu
12     public partial class FarmsForm : Form
13     {
14         string connString = "Host=ep-bold-surf-apr6yz3.c-7.us-east-1.amazonaws.com;Database=neondb;Username=neondb_owner;Password=npq_nqxusDFFP10g;SSL Mode=Require;Trust Server Certificate=true;";
15         public FarmsForm()
16         {
17             InitializeComponent();
18             this.Load += FarmsForm_Load;
19             btnAdd.Click += btnAdd_Click;
20             btnUpdate.Click += btnUpdate_Click;
21             btnDelete.Click += btnDelete_Click;
22             btnBack.Click += btnBack_Click;
23             dataGridView1.CellClick += dataGridView1_CellClick;
24         }
25     }
26
27     // LOAD
28     4 baguruu
29     void LoadFarms()
30     {
31         using (var conn = new NpgsqlConnection(connString))
32         {
33             conn.Open();
34
35             string query = "SELECT * FROM farms";
36
37             using (var da = new NpgsqlDataAdapter(query, conn))
38             {
39                 DataTable dt = new DataTable();
40                 da.Fill(dt);
41                 dataGridView1.DataSource = dt;
42             }
43         }
44     }
45
46     1 baguruu
47     private void FarmsForm_Load(object sender, EventArgs e)
48     {
49         LoadFarms();
50     }
51 }
```

```
FarmsForm.cs X
SmartFarmingSystem SmartFarmingSystem.FarmsForm connString
51 // ADD
52 private void btnAdd_Click(object sender, EventArgs e)
53 {
54     if (txtFarmName.Text == "" || txtLocation.Text == "" || txtSize.Text == "")
55     {
56         MessageBox.Show("Boş alan bırakma!");
57         return;
58     }
59
60     using (var conn = new NpgsqlConnection(connString))
61     {
62         conn.Open();
63
64         string query = "INSERT INTO farms (farm_name, location, size_acre) VALUES (@n, @l, @s)";
65
66         using (var cmd = new NpgsqlCommand(query, conn))
67         {
68             cmd.Parameters.AddWithValue("@n", txtFarmName.Text);
69             cmd.Parameters.AddWithValue("@l", txtLocation.Text);
70             cmd.Parameters.AddWithValue("@s", int.Parse(txtSize.Text));
71
72             try
73             {
74                 cmd.ExecuteNonQuery();
75             }
76             catch
77             {
78                 MessageBox.Show("Bu farm zaten var!");
79             }
80         }
81     }
82
83     LoadFarms();
84 }
85
86 // UPDATE
87 private void btnUpdate_Click(object sender, EventArgs e)
88 {
89     if (dataGridView1.CurrentRow != null)
90     {
91         int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
92
93         using (var conn = new NpgsqlConnection(connString))
94         {
95             conn.Open();
96
97             string query = @"UPDATE farms
98                             SET farm_name=@n, location=@l, size_acre=@s
99                             WHERE id=@id";
100         }
101     }
102 }
```

```
FarmsForm.cs
SmartFarmingSystem
SmartFarmingSystem.FarmsForm
connString

84
85 // UPDATE
86 1 backup
87 private void btnUpdate_Click(object sender, EventArgs e)
88 {
89     if (dataGridView1.CurrentRow != null)
90     {
91         int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
92         using (var conn = new NpgsqlConnection(connString))
93         {
94             conn.Open();
95
96             string query = @"UPDATE farms
97                             SET farm_name=@n, location=@l, size_acre=@s
98                             WHERE farm_id=@id";
99
100             using (var cmd = new NpgsqlCommand(query, conn))
101             {
102                 cmd.Parameters.AddWithValue("@n", txtFarmName.Text);
103                 cmd.Parameters.AddWithValue("@l", txtLocation.Text);
104                 cmd.Parameters.AddWithValue("@s", int.Parse(txtSize.Text));
105                 cmd.Parameters.AddWithValue("@id", id);
106
107                 cmd.ExecuteNonQuery();
108             }
109         }
110         LoadFarms();
111     }
112 }
113
114 // DELETE
115 1 backup
116 private void btnDelete_Click(object sender, EventArgs e)
117 {
118     if (dataGridView1.CurrentRow != null)
119     {
120         int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
121         using (var conn = new NpgsqlConnection(connString))
122         {
123             conn.Open();
124
125             string query = "DELETE FROM farms WHERE farm_id=@id";
126
127             using (var cmd = new NpgsqlCommand(query, conn))
128             {
129                 cmd.Parameters.AddWithValue("@id", id);
130                 cmd.ExecuteNonQuery();
131             }
132         }
133     }
134 }
```

```
FarmsForm.cs
SmartFarmingSystem
SmartFarmingSystem.FarmsForm
connString

131         cmd.ExecuteNonQuery();
132     }
133 }
134 LoadFarms();
135 }
136 }
137 }
138
139 // GRID CLICK
140 private void dataGridView1_CellClick(object sender, DataGridViewCellEventArgs e)
141 {
142     if (e.RowIndex >= 0)
143     {
144         var row = dataGridView1.Rows[e.RowIndex];
145
146         txtFarmName.Text = row.Cells[1].Value.ToString();
147         txtLocation.Text = row.Cells[2].Value.ToString();
148         txtSize.Text = row.Cells[3].Value.ToString();
149     }
150 }
151
152 // BACK
153 private void btnBack_Click(object sender, EventArgs e)
154 {
155     this.Hide();
156     Dashboard d = new Dashboard();
157     d.Show();
158     this.Close();
159 }
160
161 private void label7_Click(object sender, EventArgs e)
162 {
163 }
164
165 private void label2_Click(object sender, EventArgs e)
166 {
167 }
168
169 }
170
171
172
173
174 }
```


Sensor management

Adding sensor

SensorReadingsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

SENSOR MANAGEMENT

Temperature:

Moisture:

Field:

▼

Date: 14 Mayis 2026 Perşembe

📅

ADD

UPDATE

DELETE

BACK

	reading_id	field_name	temperature	moisture	reading_date
▶	1	Greenhouse A	22	70	6.05.2026 10:00
	2	Greenhouse B	22	70	6.05.2026 10:00
	3	Vertical Tower 1	23	54	14.05.2026 16:09
	16	Greenhouse B	24	80	14.05.2026 16:44
*					

Updating sensor

SensorReadingsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

SENSOR MANAGEMENT

Temperature: 24

Moisture: 80

Field:

Greenhouse B ▼

Date: 14 Mayis 2026 Perşembe

📅

ADD

UPDATE

DELETE

BACK

	reading_id	field_name	temperature	moisture	reading_date
▶	1	test field	22	70	6.05.2026 10:00
	2	Greenhouse B	22	70	6.05.2026 10:00
	3	Vertical Tower 1	23	54	14.05.2026 16:09
	16	Greenhouse B	24	80	14.05.2026 16:44
*					

Deleting sensor

SensorReadingsForm

STRAWBERRY FARMING CORP

Welcome! Have a nice day

SENSOR MANAGEMENT

Temperature: 24

Moisture: 80

Field: Greenhouse B

Date: 14 Mayıs 2026 Perşembe

ADD UPDATE DELETE

BACK

	reading_id	field_name	temperature	moisture	reading_date
▶	1	test field	22	70	6.05.2026 10:00
	2	Greenhouse B	22	70	6.05.2026 10:00
	3	Vertical Tower 1	23	54	14.05.2026 16:09
*					

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Data;
5 using System.Drawing;
6 using System.Linq;
7 using System.Windows.Forms;
8 using Npgsql;
9
10 namespace SmartFarmingSystem
11 {
12     3 bayyuru
13     public partial class SensorReadingsForm : Form
14     {
15         int selectedReadingId = 0;
16         string connString = "Host=ep-bold-surf-apre6yz3.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npq_nqxusDFfP1Bg;SSL Mode=Require;Trust Server Certificate=true;";
17
18         4 bayyuru
19         void LoadSensors()
20         {
21             using (var conn = new NpgsqlConnection(connString))
22             {
23                 conn.Open();
24
25                 string query = @"SELECT s.reading_id, f.field_name, s.temperature, s.moisture, s.reading_date
26                     FROM sensorreadings s
27                     JOIN fields f ON s.field_id = f.field_id";
28
29                 using (var da = new NpgsqlDataAdapter(query, conn))
30                 {
31                     DataTable dt = new DataTable();
32                     da.Fill(dt);
33                     dataGridView1.DataSource = dt;
34                 }
35             }
36
37             1 bayyuru
38             void LoadFields()
39             {
40                 using (var conn = new NpgsqlConnection(connString))
41                 {
42                     conn.Open();
43
44                     string q = "SELECT field_id, field_name FROM fields";
45
46                     using (var da = new NpgsqlDataAdapter(q, conn))
47                     {
48                         DataTable dt = new DataTable();
49                         da.Fill(dt);
50                         cmbField.DataSource = dt;
51                     }
52                 }
53             }
54         }
55     }
56 }
```

```
SensorReadingsForm.cs
SmartFarmingSystem
SmartFarmingSystem.SensorReadingsForm
LoadFields()

47
48     cmbField.DataSource = dt;
49     cmbField.DisplayMember = "field_name"; // görünen
50     cmbField.ValueMember = "field_id";     // arka plan (ID)
51 }
52
53
54
55 private void SensorReadingsForm_Load(object sender, EventArgs e)
56 {
57     LoadSensors();
58     LoadFields();
59 }
60
61 public SensorReadingsForm()
62 {
63     InitializeComponent();
64     this.Load += SensorReadingsForm_Load;
65     this.StartPosition = FormStartPosition.CenterScreen;
66     this.Size = new Size(1000, 600);
67     this.FormBorderStyle = FormBorderStyle.FixedSingle;
68     this.MaximizeBox = false;
69 }
70
71 private void btnAdd_Click(object sender, EventArgs e)
72 {
73     using (var conn = new NpgsqlConnection(connString))
74     {
75         conn.Open();
76
77         string query = @"INSERT INTO sensorreadings (field_id, temperature, moisture, reading_date)
78             VALUES (
79                 (SELECT field_id FROM fields WHERE field_name=@f),
80                 @temp, @moist, @date
81             )";
82
83         using (var cmd = new NpgsqlCommand(query, conn))
84         {
85             cmd.Parameters.AddWithValue("@f", cmbField.Text);
86             cmd.Parameters.AddWithValue("@temp", int.Parse(txtTemperature.Text));
87             cmd.Parameters.AddWithValue("@moist", int.Parse(txtMoisture.Text));
88             cmd.Parameters.AddWithValue("@date", dtReadingDate.Value);
89
90             cmd.ExecuteNonQuery();
91         }
92     }
93
94     LoadSensors();
95     txtTemperature.Clear();
96 }
```

```
SensorReadingsForm.cs
SmartFarmingSystem
SmartFarmingSystem.SensorReadingsForm
LoadFields()

93
94     LoadSensors();
95     txtTemperature.Clear();
96     txtMoisture.Clear();
97     cmbField.SelectedIndex = -1;
98 }
99
100 private void btnDelete_Click(object sender, EventArgs e)
101 {
102     if (dataGridView1.CurrentRow != null)
103     {
104         int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells["reading_id"].Value);
105
106         using (var conn = new NpgsqlConnection(connString))
107         {
108             conn.Open();
109
110             string query = "DELETE FROM sensorreadings WHERE reading_id=@id";
111
112             using (var cmd = new NpgsqlCommand(query, conn))
113             {
114                 cmd.Parameters.AddWithValue("@id", id);
115                 cmd.ExecuteNonQuery();
116             }
117         }
118
119         LoadSensors();
120     }
121 }
122
123 private void btnUpdate_Click(object sender, EventArgs e)
124 {
125     if (selectedReadingId == 0)
126     {
127         MessageBox.Show("Lütfen bir kayıt seç!");
128         return;
129     }
130
131     if (cmbField.SelectedValue == null)
132     {
133         MessageBox.Show("Field seçili değil!");
134         return;
135     }
136
137     using (var conn = new NpgsqlConnection(connString))
138     {
139         conn.Open();
140
141         string q = @"UPDATE sensorreadings
142             SET field_id = @field_id, temperature = @temp, moisture = @moist, reading_date = @date
143             WHERE reading_id = @id";
144
145         using (var cmd = new NpgsqlCommand(q, conn))
146         {
147             cmd.Parameters.AddWithValue("@field_id", cmbField.SelectedValue);
148             cmd.Parameters.AddWithValue("@temp", int.Parse(txtTemperature.Text));
149             cmd.Parameters.AddWithValue("@moist", int.Parse(txtMoisture.Text));
150             cmd.Parameters.AddWithValue("@date", dtReadingDate.Value);
151             cmd.Parameters.AddWithValue("@id", selectedReadingId);
152
153             cmd.ExecuteNonQuery();
154         }
155     }
156
157     LoadSensors();
158 }
```

```
SensorReadingsForm.cs
SmartFarmingSystem
SmartFarmingSystem.SensorReadingsForm
LoadFields()

136
137
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144
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183
184
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188

using (var conn = new NpgsqlConnection(connString))
{
    conn.Open();

    string q = @"UPDATE sensorreadings
    SET temperature=@temp,
        moisture=@moisture,
        field_id=@field_id,
        reading_date=@date
    WHERE reading_id=@id";

    using (var cmd = new NpgsqlCommand(q, conn))
    {
        cmd.Parameters.AddWithValue("@temp", Convert.ToDouble(txtTemperature.Text));
        cmd.Parameters.AddWithValue("@moisture", Convert.ToDouble(txtMoisture.Text));
        cmd.Parameters.AddWithValue("@field_id", Convert.ToInt32(cmbField.SelectedValue)); // FIX
        cmd.Parameters.AddWithValue("@date", dtReadingDate.Value);
        cmd.Parameters.AddWithValue("@id", selectedReadingId);

        cmd.ExecuteNonQuery();
    }

    LoadSensors();
}

1 baguru
private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs e)
{
    if (e.RowIndex >= 0)
    {
        var row = dataGridView1.Rows[e.RowIndex]; // BURASI $ART

        if (row.Cells[0].Value != DBNull.Value)
            selectedReadingId = Convert.ToInt32(row.Cells[0].Value);

        txtTemperature.Text = row.Cells["temperature"].Value?.ToString();
        txtMoisture.Text = row.Cells["moisture"].Value?.ToString();
        cmbField.Text = row.Cells["field_name"].Value?.ToString();

        var v = row.Cells["reading_date"].Value;
        if (v != DBNull.Value && v != null)
            dtReadingDate.Value = Convert.ToDateTime(v);
    }
}

1 baguru
private void btnBack_Click(object sender, EventArgs e)
{
    Dashboard d = new Dashboard();
}
```

Fields Management

Adding Field

FieldsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

 FIELDS MANAGEMENT

Farm: Smart Berry Lab

Field Name: test field2

Soil: Hydroponic2

Area: 55

ADDUPDATEDELETE

BACK

	field_id	farm_name	field_name	soil_type	area
▶	1	Hydro Strawber...	Greenhouse A	Hydroponic	3
	2	Hydro Strawber...	Greenhouse B	Hydroponic	4
	3	Smart Berry Lab	Vertical Tower 1	Hydroponic	2
	11	Smart Berry Lab	test field2	Hydroponic2	55
*					

Updating Field

FieldsForm

 STRAWBERRY FARMING CORP

Welcome! Have a nice day

 FIELDS MANAGEMENT

Farm: Smart Berry Lab

Field Name: test field 11

Soil: Hydroponic22

Area: 5

ADDUPDATEDELETE

BACK

	field_id	farm_name	field_name	soil_type	area
▶	2	Hydro Strawber...	Greenhouse B	Hydroponic	4
	3	Smart Berry Lab	Vertical Tower 1	Hydroponic	2
	1	Smart Berry Lab	Greenhouse A	Hydroponic	3
	11	Smart Berry Lab	test field 11	Hydroponic22	5
*					

Deleting Field

FieldsForm



STRAWBERRY FARMING CORP

Welcome! Have a nice day

 FIELDS MANAGEMENT

Farm:Smart Berry Lab

Field Name:test field 11

Soil:Hydroponic22

Area:5

ADDUPDATEDELETE

BACK

	field_id	farm_name	field_name	soil_type	area
▶	2	Hydro Strawber...	Greenhouse B	Hydroponic	4
	3	Smart Berry Lab	Vertical Tower 1	Hydroponic	2
	1	Smart Berry Lab	Greenhouse A	Hydroponic	3
✱					

```
FieldForm.cs
SmartFarmingSystem
using Npgsql;
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Text;
using System.Windows.Forms;

namespace SmartFarmingSystem
{
    public partial class FieldsForm : Form
    {
        string connString = "Host=ep-bold-surf-apre5y23.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npg_nqxUsDffP10g;SSL Mode=Require;Trust Server Certificate=true;";

        public FieldsForm()
        {
            InitializeComponent();
            this.Load += FieldsForm_Load;
            this.StartPosition = FormStartPosition.CenterScreen;

            this.FormBorderStyle = FormBorderStyle.FixedSingle;
            this.MaximizeBox = false;
            btnAdd.Click += btnAdd_Click;
            btnUpdate.Click += btnUpdate_Click;
            btnDelete.Click += btnDelete_Click;
            btnBack.Click += btnBack_Click;
            dataGridView1.CellClick += dataGridView1_CellClick;
        }

        // LOAD FARMS → dropdown
        void LoadFarms()
        {
            using (var conn = new NpgsqlConnection(connString))
            {
                conn.Open();

                string query = "SELECT farm_id, farm_name FROM farms";

                using (var da = new NpgsqlDataAdapter(query, conn))
                {
                    DataTable dt = new DataTable();
                    da.Fill(dt);

                    cmbFarm.DataSource = dt;
                    cmbFarm.DisplayMember = "farm_name";
                    cmbFarm.ValueMember = "farm_id";
                }
            }
        }
    }
}
```

```
FieldForm.cs
SmartFarmingSystem
// LOAD FIELDS
4 bapuru
void LoadFields()
{
    using (var conn = new NpgsqlConnection(connString))
    {
        conn.Open();

        string query = @"SELECT f.field_id, fa.farm_name, f.field_name, f.soil_type, f.area
                        FROM fields f
                        JOIN farms fa ON f.farm_id = fa.farm_id";

        using (var da = new NpgsqlDataAdapter(query, conn))
        {
            DataTable dt = new DataTable();
            da.Fill(dt);
            dataGridView1.DataSource = dt;
        }
    }
}

1 bapuru
private void FieldsForm_Load(object sender, EventArgs e)
{
    LoadFarms();
    LoadFields();
}

// + ADD
1 bapuru
private void btnAdd_Click(object sender, EventArgs e)
{
    if (txtFieldName.Text == "")
    {
        MessageBox.Show("Boş alan bırakma!");
        return;
    }

    using (var conn = new NpgsqlConnection(connString))
    {
        conn.Open();

        string query = @"INSERT INTO fields (farm_id, field_name, soil_type, area)
                        VALUES (@farm, @name, @soil, @area)";

        using (var cmd = new NpgsqlCommand(query, conn))
        {
            cmd.Parameters.AddWithValue("@farm", cmbFarm.SelectedValue);
            cmd.Parameters.AddWithValue("@name", txtFieldName.Text);
            cmd.Parameters.AddWithValue("@soil", txtSoil.Text);
        }
    }
}
```

```

FieldForm.cs
SmartFarmingSystem
SmartFarmingSystem.FieldForm
label2_Click(object sender, EventArgs e)

97      cmd.Parameters.AddWithValue("@name", txtFieldName.Text);
98      cmd.Parameters.AddWithValue("@soil", txtSoil.Text);
99      cmd.Parameters.AddWithValue("@area", int.Parse(txtArea.Text));
100
101      cmd.ExecuteNonQuery();
102  }
103  }
104
105      LoadFields();
106  }
107
108  // UPDATE
109  private void btnUpdate_Click(object sender, EventArgs e)
110  {
111      if (dataGridView1.CurrentRow != null)
112      {
113          int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
114
115          using (var conn = new NpgsqlConnection(connString))
116          {
117              conn.Open();
118
119              string query = @"UPDATE fields
120                             SET farm_id=@farm,
121                                 field_name=@name,
122                                 soil_type=@soil,
123                                 area=@area
124                             WHERE field_id=@id";
125
126              using (var cmd = new NpgsqlCommand(query, conn))
127              {
128                  cmd.Parameters.AddWithValue("@farm", cmbFarm.SelectedValue);
129                  cmd.Parameters.AddWithValue("@name", txtFieldName.Text);
130                  cmd.Parameters.AddWithValue("@soil", txtSoil.Text);
131                  cmd.Parameters.AddWithValue("@area", int.Parse(txtArea.Text));
132                  cmd.Parameters.AddWithValue("@id", id);
133
134                  cmd.ExecuteNonQuery();
135              }
136
137              LoadFields();
138          }
139      }
140  }
141
142  // DELETE
143  private void btnDelete_Click(object sender, EventArgs e)
144  {
145      if (dataGridView1.CurrentRow != null)

```

```

FieldForm.cs
SmartFarmingSystem
SmartFarmingSystem.FieldForm
label2_Click(object sender, EventArgs e)

141
142  // DELETE
143  private void btnDelete_Click(object sender, EventArgs e)
144  {
145      if (dataGridView1.CurrentRow != null)
146      {
147          int id = Convert.ToInt32(dataGridView1.CurrentRow.Cells[0].Value);
148
149          using (var conn = new NpgsqlConnection(connString))
150          {
151              conn.Open();
152
153              string query = "DELETE FROM fields WHERE field_id=@id";
154
155              using (var cmd = new NpgsqlCommand(query, conn))
156              {
157                  cmd.Parameters.AddWithValue("@id", id);
158                  cmd.ExecuteNonQuery();
159              }
160
161              LoadFields();
162          }
163      }
164  }
165
166  // GRID CLICK
167  private void dataGridView1_CellClick(object sender, DataGridViewCellEventArgs e)
168  {
169      if (e.RowIndex >= 0)
170      {
171          var row = dataGridView1.Rows[e.RowIndex];
172
173          cmbFarm.Text = row.Cells[1].Value.ToString();
174          txtFieldName.Text = row.Cells[2].Value.ToString();
175          txtSoil.Text = row.Cells[3].Value.ToString();
176          txtArea.Text = row.Cells[4].Value.ToString();
177      }
178  }
179
180  // BACK
181  private void btnBack_Click(object sender, EventArgs e)
182  {
183      this.Hide();
184      Dashboard d = new Dashboard();
185      d.Show();
186      this.Close();
187  }
188

```


Statistics Form

Gui screenshots of statistics are given on DML part. Statistics form codes:

```
StatisticsForm.cs
SmartFarmingSystem
SmartFarmingSystem.StatisticsForm
btnQ4_Click(object sender, EventArgs e)

1 using Npgsql;
2 using System;
3 using System.Collections.Generic;
4 using System.ComponentModel;
5 using System.Data;
6 using System.Drawing;
7 using System.Text;
8 using System.Windows.Forms;
9
10 namespace SmartFarmingSystem
11 {
12     5 bayyuru
13     public partial class StatisticsForm : Form
14     {
15         string connString = "Host=ep-bold-surf-apr6yz3.c-7.us-east-1.aws.neon.tech;Database=neondb;Username=neondb_owner;Password=npqUsDFP10g;SSL Mode=Require;Trust Server Certificate=true;";
16         int selectedCropId = -1;
17         1 bayyuru
18         public StatisticsForm()
19         {
20             InitializeComponent();
21         }
22
23         6 bayyuru
24         void LoadData(string query)
25         {
26             using (var conn = new NpgsqlConnection(connString))
27             {
28                 conn.Open();
29                 using (var da = new NpgsqlDataAdapter(query, conn))
30                 {
31                     DataTable dt = new DataTable();
32                     da.Fill(dt);
33                     dataGridView1.DataSource = dt;
34                 }
35             }
36
37             //This query shows fields with an average temperature higher than the overall system average.
38             1 bayyuru
39             private void btnQ1_Click(object sender, EventArgs e)
40             {
41                 string q = @"
42                 SELECT f.field_name, AVG(s.temperature) AS avg_temp
43                 FROM sensorreadings s
44                 JOIN fields f ON s.field_id = f.field_id
45                 GROUP BY f.field_name
46                 HAVING AVG(s.temperature) > (
47                     SELECT AVG(temperature) FROM sensorreadings
48                 );";
49
50                 LoadData(q);
51             }
52
53             // o This query calculates the total number of crops in each farm using multiple table joins.
54             1 bayyuru
55             private void btnQ2_Click(object sender, EventArgs e)
56             {
57                 string q = @"
58                 SELECT fa.farm_name, COUNT(c.crop_id) AS total_crops
59                 FROM farms fa
60                 JOIN fields f ON fa.farm_id = f.farm_id
61                 JOIN crops c ON f.field_id = c.field_id
62                 GROUP BY fa.farm_name
63                 ORDER BY total_crops DESC";
64                 LoadData(q);
65             }
66
67             // o This query displays the minimum and maximum temperature recorded in each field.
68             1 bayyuru
69             private void btnQ3_Click(object sender, EventArgs e)
70             {
71                 string q = @"
72                 SELECT f.field_name,
73                        MAX(s.temperature) AS max_temp,
74                        MIN(s.temperature) AS min_temp
75                 FROM sensorreadings s
76                 JOIN fields f ON s.field_id = f.field_id
77                 GROUP BY f.field_name";
78                 LoadData(q);
79             }
80
81             // o This query identifies fields that frequently experience low moisture levels.
82             1 bayyuru
83             private void btnQ4_Click(object sender, EventArgs e)
84             {
85                 string q = @"
86                 SELECT f.field_name,
87                        MAX(s.moisture) AS highest_moisture
88                 FROM fields f
89                 JOIN sensorreadings s
90                 ON f.field_id = s.field_id
91                 GROUP BY f.field_name
92                 ORDER BY highest_moisture DESC
93                 LIMIT 1;";
94             }
95         }
96     }
97 }
```



```
StatisticsForm.cs x
SmartFarmingSystem SmartFarmingSystem.StatisticsForm btnQ4_Click(object sender, EventArgs e)
89 #
90 GROUP BY f.field_name
91 ORDER BY highest_moisture DESC
92 LIMIT 1;
93 ";
94 LoadData(q);
95 }
96
97
98
99 // o This query retrieves the most recent sensor reading for each field.
100 1 basipuru
101 private void btnQ5_Click(object sender, EventArgs e)
102 {
103     string q = @"
104     SELECT f.field_name, s.temperature, s.reading_date
105     FROM sensorreadings s
106     JOIN fields f ON s.field_id = f.field_id
107     WHERE s.reading_date = (
108         SELECT MAX(s2.reading_date)
109         FROM sensorreadings s2
110         WHERE s2.field_id = s.field_id
111     )";
112     LoadData(q);
113 }
114
115 // o This query lists fields that currently have no crops assigned.
116 1 basipuru
117 private void btnQ6_Click(object sender, EventArgs e)
118 {
119     string q = @"
120     SELECT f.field_name
121     FROM fields f
122     LEFT JOIN crops c ON f.field_id = c.field_id
123     WHERE c.crop_id IS NULL";
124     LoadData(q);
125 }
126
127 1 basipuru
128 private void btnBack_Click(object sender, EventArgs e)
129 {
130     Dashboard d = new Dashboard();
131     d.Show();
132     this.Close();
133 }
```

6) GitHub Link:

[Strawberry farming corp.](#)